

interest with a single touch.

The analyser is designed to make both major and minor upgrades practical, reducing the chances of equipment becoming obsolete. The upgradable features include real-time spectrum and vector signal analysis, frequency mask triggering, internal preamplifiers and external mixing. The analyser computer infrastructure can also be updated to include faster processors, larger or solid-state disk drives, and faster data interfaces. These upgrades are especially helpful for processor- and data-intensive applications such as digital demodulation and vector signal analysis.

Now let us take a look at some of the other launches in this segment.

High-performance UHD TV testing

Broadcast Video Expo (BVE) London 2016 saw the PHABRIX Qx featuring field-programmable gate array (FPGA) technology, which is capable of ultra-high-definition television (UHD TV) testing. Separately, UK based Omnitek also announced that they were shipping a number of pre-ordered UHD Ultra 4K Tool Box units, with the world's first 12G-SDI test technology, to early adopters.

The PHABRIX Qx provides generation and analysis tools on signals with up to 48Gbps payloads offering both video and audio support. Incorporating real-time 12G serial digital interface (SDI) physical layer analysis (eye and jitter) tool-set with automatic Society of Motion Picture & Television Engineers (SMPTE) compliance measurements, it provides comprehensive testing for both UHD TV1 and UHD TV2 infrastructures. Phil Adams, managing director, PHABRIX, commented, "Despite the technological challenges of making a true hybrid solution for SDI and IP infrastructure, the implementation of the Qx enables us

to respond to the needs of customers now and in the future."

Other releases

A new RSA500 and RSA600 series of analysers from Tektronix offers frequency coverage from 9kHz to 7.5GHz with 40MHz acquisition bandwidth, a measurement dynamic range from -161dBm/Hz displayed average noise level and up to +30dBm maximum input. These USB based real-time spectrum ana-



Tektronix RSA 500 series

lysers have four new models targeting design, spectrum management, wireless transmitter installation and maintenance applications. Using a USB 3.0 connection, the instruments can be operated via a desktop PC, laptop or tablet using a Signal Vu-PC software suite.

IneoQuest and SIGOS demonstrated a prototype system at the 2016 Mobile World Congress in Barcelona. The solution combines IneoQuest's IQ Dialogue software with the SIGOS SITE Test Systems' remote test units, creating an integrated solution.

In an announcement, Adil Kaya, CEO, SIGOS said, "In the mobile telecommunications industry, video was always a topic; getting reliable quality and performance indicators for mobile video services has been a challenge. For over 400 network operators the SITE architecture, meas-

urements and results are a highly trusted source to test and measure quality - adding video is just a logical consequence." This can be used to conduct active testing and evaluation of adaptive bit rate (ABR) video streams in all major streaming formats, including Hypertext Transfer Protocol (HTTP) live streaming (HLS), HTTP dynamic streaming (HDS), Microsoft smooth streaming (MSS) and dynamic adaptive streaming over HTTP (DASH)—and across any video-capable access network technology, including 3G, LTE, LAN, DSL and Wi-Fi.

The market

The major buyers of audio-video T&M equipment are broadcast service providers, R&D labs, product manufacturers and local ISP vendors. With the race for bigger and better displays and acoustics, manufacturers of smartphones and their accessories have become one of the biggest patrons of testing equipment. This might have led to a marginal price increase in the products for end users, but the quality has significantly improved, which is a minor trade-off.

Test equipment for audio-video parameters has been in the market for quite some time now. And the boom in the media and broadcasting industry has given it a very wide market to cater to. The changes with every new launch are taking the audio/video quality to further new heights. We use technology on a daily basis, without wondering how much research goes into the making and proper maintenance of such systems. Next time you use your smartphone, you might wonder how balanced the acoustics are, or how the development team is able to efficiently integrate more than 400ppi pixel density in a small screen. **EFY**